Python Engineering - Class 07 Theory Guide

NumPy Arrays

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Array geometry

NumPy separates the physical meaning of data from the structural dimensions used to store it.

Relationship / equation

```
3D physical vector [x,y,z] can have shape (3,) and ndim=1
```

Connection to code

ndim counts axes in the data structure; it does not mean spatial dimension.

2. Vectorization

An operation written once can be applied to all array elements without an explicit Python loop.

Relationship / equation

```
energy = g*h + 0.5*v**2
```

Connection to code

If h and v are columns with N samples, the result is an N-element energy vector.

3. Shape compatibility

Many NumPy operations require compatible shapes. Broadcasting compares dimensions from the right.

Relationship / equation

```
(N,3) with (3,) is often compatible
```

Connection to code

Understanding shape before coding prevents many silent or confusing numerical mistakes.

4. Axes

axis identifies a structural direction in an array.

Relationship / equation

```
axis=0 reduces rows; axis=1 reduces columns for a 2D dataset
```

Connection to code

For shape (N,3), reducing axis=0 leaves one result per column.

5. Dynamic growth vs numerical storage

NumPy arrays are fixed-size blocks. Repeated vstack in a large loop reallocates/copies data.

Relationship / equation

```
unknown N -> list append then np.array; known N -> preallocate
```

Connection to code

The course initially uses small arrays for clarity, then introduces more efficient patterns.

Manual validation method

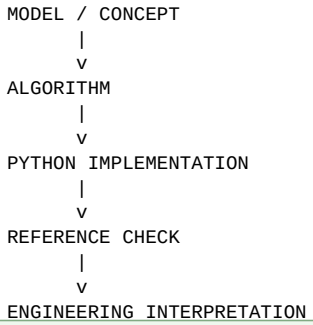
For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

NumPy arrays become the standard representation for numerical methods, vectors, matrices and engineering datasets.

Final summary



> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.